



Systematic Layout Planning (SLP)

March 2026
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What is a facility:

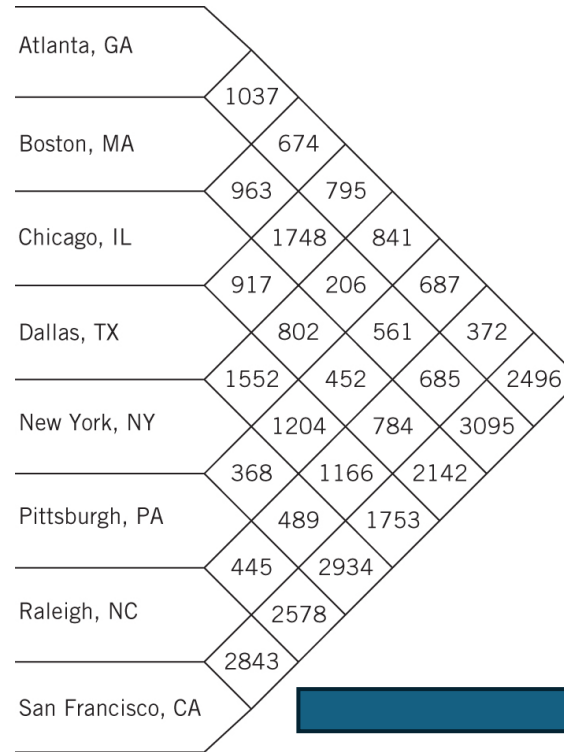
A FACILITY includes all fixed assets to achieve a production/service objective

- Design year = 5-10 years into future
- Implying lots of uncertainty, designers inflate estimates to 'hedge their bets'
- Major stages in factory include: Supply (purchasing), manufacture (production/operations) and distribution (sales)
- Overall flow from raw materials to finished goods = logistics or supply chain management system
- Input (receiving) and Output (dispatch) – this can happen for whole facility or between 'departments' or even workstations (input/pickup/ingress to output/delivery/egress)
- Definition of 'workstation'(3.6.1) vs 'department' (3.6.2) definition

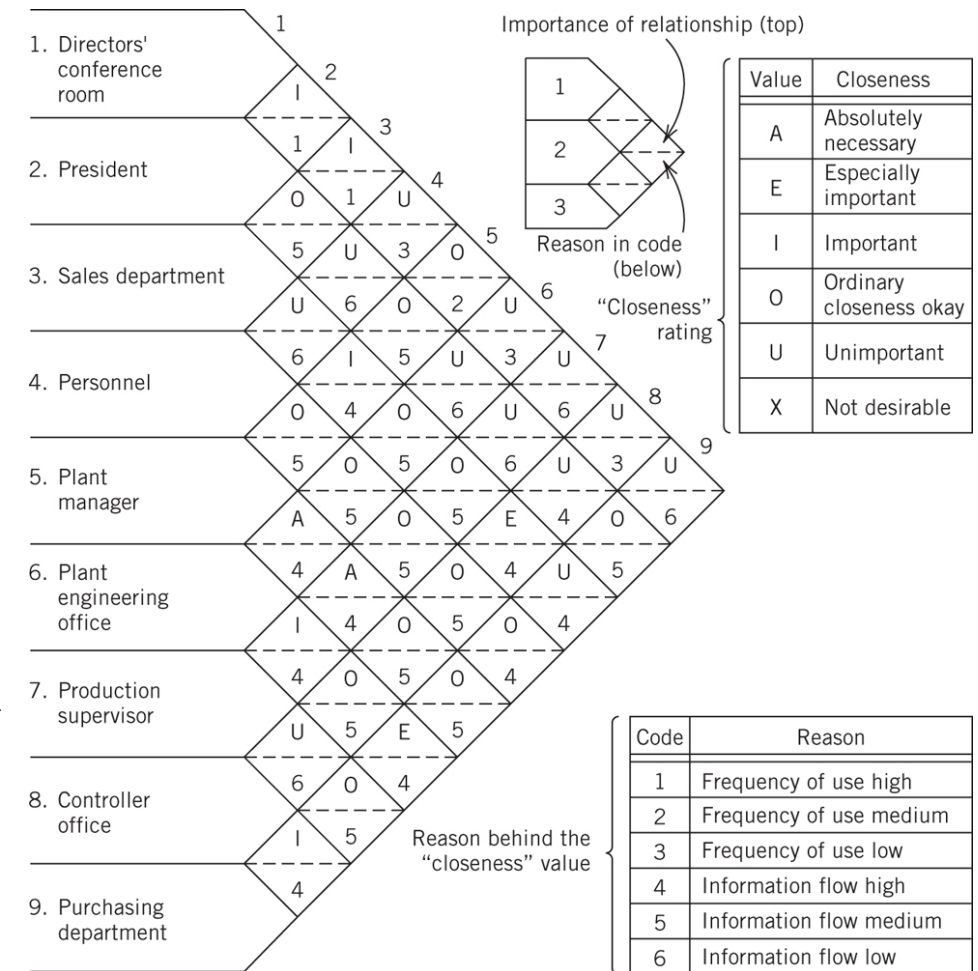
What is an Activity in a facility?

- An activity is a process or part of a process or step in a process
- Not all activities need machines, but all need functional space (e.g. meetings where strategy is planned and decisions made)
- Measuring “activities” between departments influence design e.g. physical material flow or information flow
- challenge is to define what is meant by ‘activity’ and to measure either qualitatively or quantitatively e.g. nr of movements between, if costly, then more is not effective, so aim for less)
- If idea is to minimize cost of flow:
 - Eliminating UNNECESSARY movements
 - Minimizing travel distances
 - Eliminating manual handling
 - Reducing flow density

From \ To								
	Atlanta, GA	Boston, MA	Chicago, IL	Dallas, TX	New York, NY	Pittsburgh, PA	Raleigh, NC	San Francisco, CA
Atlanta, GA		1037	674	795	841	687	372	2496
Boston, MA	1037		963	1748	206	561	685	3095
Chicago, IL	674	963		917	802	452	784	2142
Dallas, TX	795	1748	917		1552	1204	1166	1753
New York, NY	841	206	802	1552		368	489	2934
Pittsburgh, PA	687	561	452	1204	368		445	2578
Raleigh, NC	372	685	784	1166	489	445		2843
San Francisco, CA	2496	3095	2142	1753	2934	2578	2843	



If this activity includes material flow (amount moved) then From-To chart is useful as an Activity relationship chart (but this is not the only measurement)



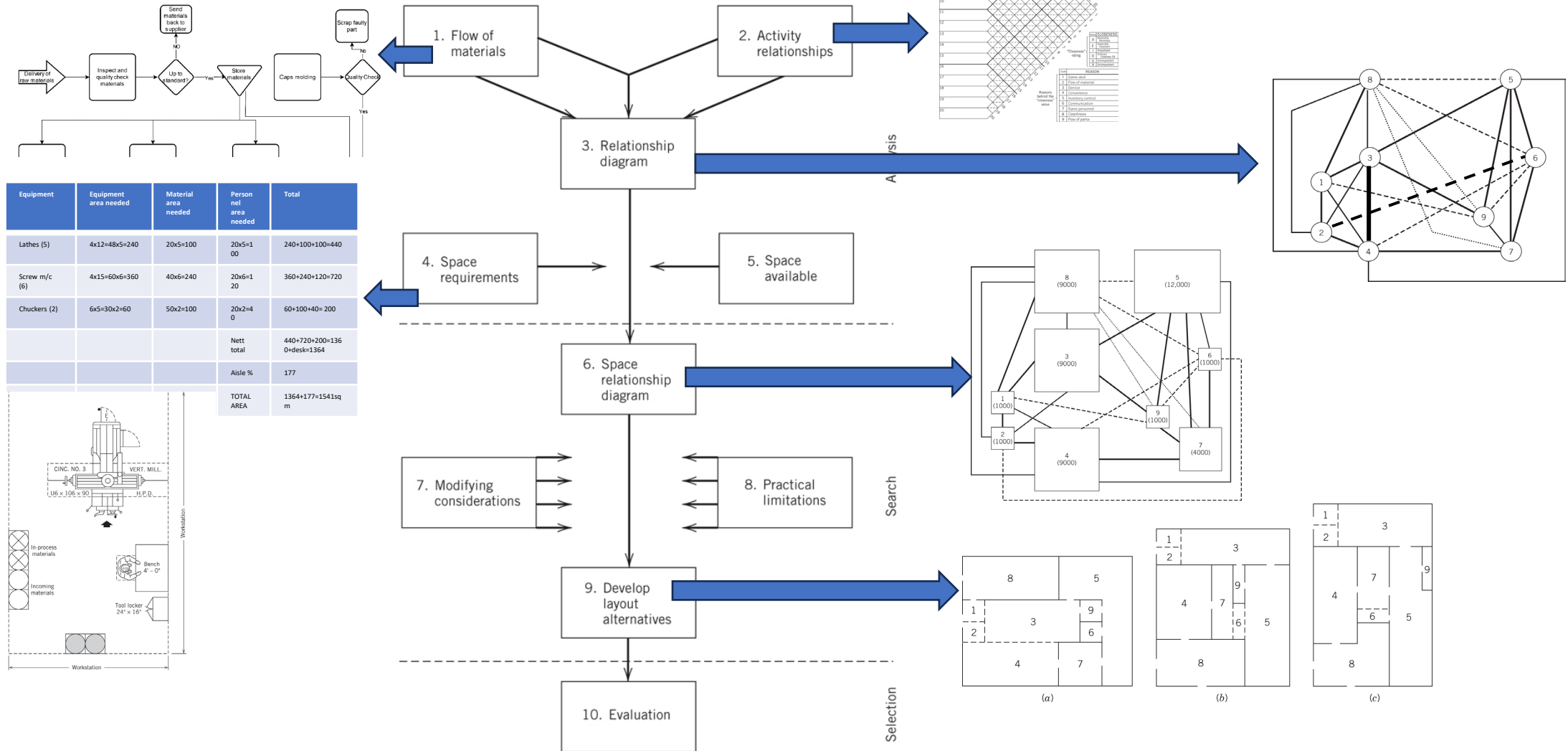
Just as distance was used between towns, flow between departments can be used to determine Closeness relationship

Definition of SLP

Jacobs & Chase : SLP is a technique for solving layout problems when the use of numerical flow data between departments is not practical. An activity relationship chart is adjusted by *trial and error* until a satisfactory adjacency pattern is obtained.

Tompkins: SLP is a procedure where block layouts are developed logically and sequentially following a step by step process and requires a good understanding of the process flow and relationships between activities.

SLP diagram logic



Planning 'departments' or functional areas or Production process clusters = key

Input data & Activities

WHAT is produced at each step in the process? (finished goods, assemblies, parts, components, raw materials needed)

HOW MUCH at each step in the process? (pieces/day, units/hour or minute etc, demand)

Activities understanding needed:

HOW or with WHAT will it be processed? (machines, operations, manual labor)

What is the SEQUENCE, logical steps? (activities in series, parallel, outsourced, produced)

HOW is it moved between steps in the process?

WHEN and HOW LONG does it take?

Understand the flow of materials

Systematic Layout Planning for a Floor of a store with 5 departments – example1 to illustrate SLP logic

A. Relationship Chart (based on Tables B and C)

From	To				Area (sq. ft.)
	2	3	4	5	
1. Credit department	I	U	E	U	100
	6	—	4	—	
2. Toy department		U	I	A	400
		—	1	1,6	
3. Wine department			U	X	300
			—	1	
4. Camera department				X	100
				1	
5. Candy department					100

Letter Closeness rating
 Number Reason for rating

- 1 A Between where is A?
- 1 E
- 2 I
- 4 U
- 2 X Between where is X's?

Rule of thumb

Code	% used
A	5%
E	10%
I	15%
O	25%

B.

CODE	REASON*
1	Type of customer
2	Ease of supervision
3	Common personnel
4	Contact necessary
5	Share same space
6	Psychology

*Others may be used.

C.

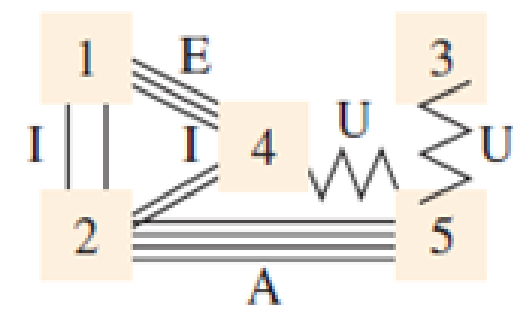
VALUE	CLOSENESS	LINE CODE*	NUMERICAL WEIGHTS
A	Absolutely necessary	=====	16
E	Especially important	=====	8
I	Important	=====	4
O	Ordinary closeness OK	=====	2
U	Unimportant	=====	0
X	Not allowed	=====	-80

*Used for example purposes only.

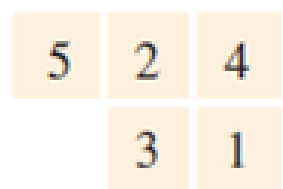
Why or when is X applicable in typical production plant?
 Noise, smell, dust, heat, cold, danger, explosion risk

Reasons are subjective e.g. movement between departments can be another logical reason based on expert opinions and input from shopfloor

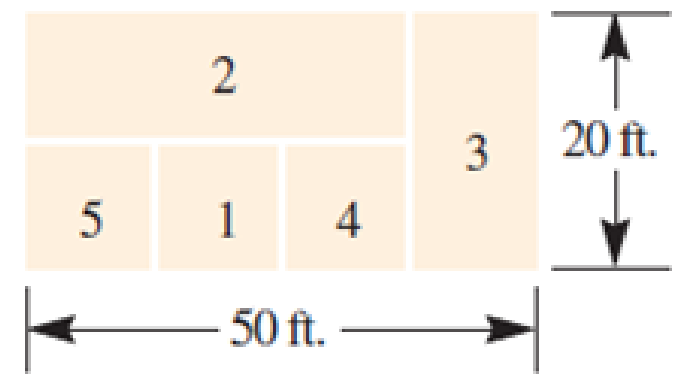
Systematic Layout Planning for a Floor of a Department Store – example to illustrate SLP logic



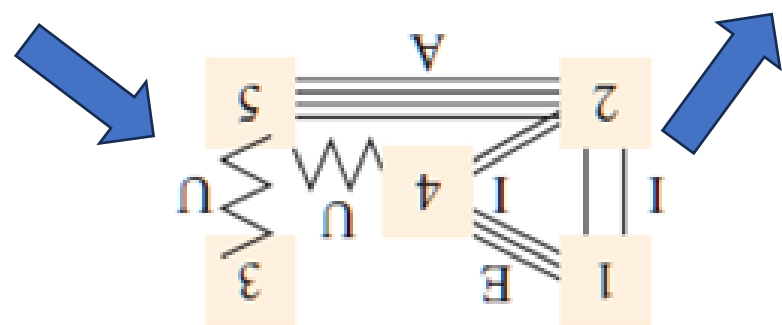
Initial relationship diagram
(based on Tables A and C)



Initial layout based on
relationship diagram
(ignoring space and
building constraints)



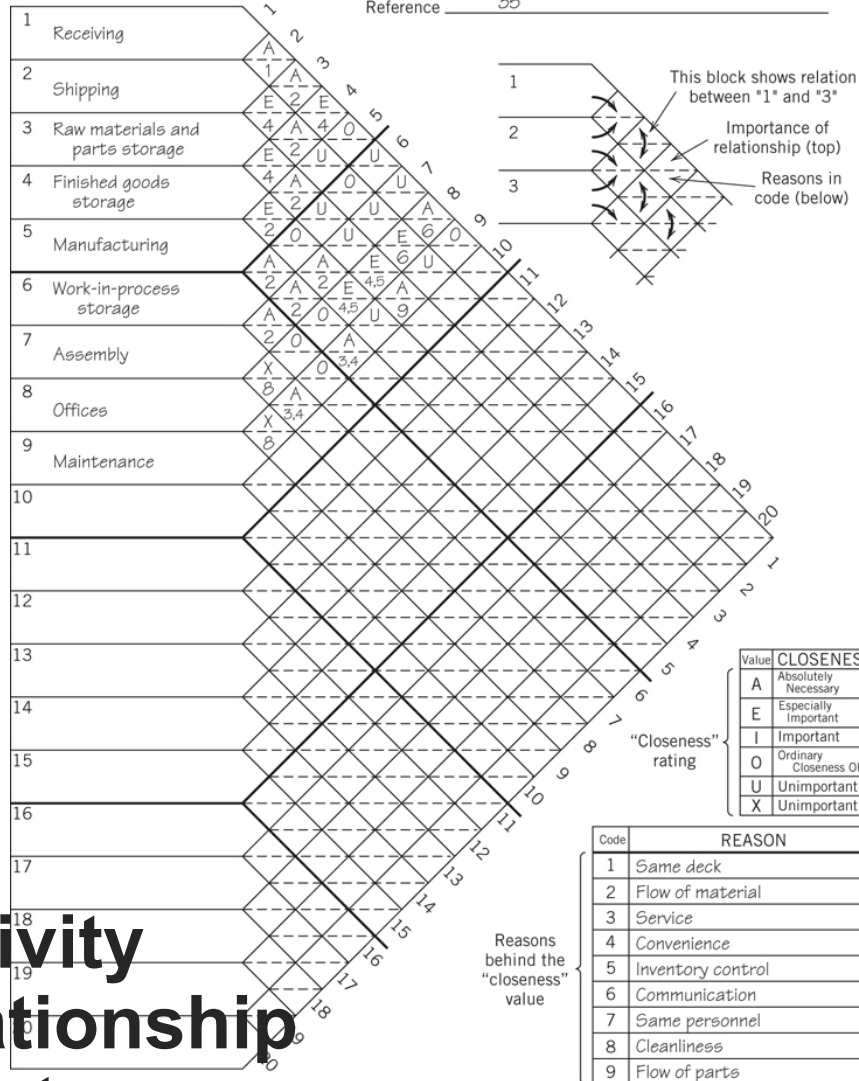
Final layout adjusted by square
footage and building size



Note that above is just a layout – other possibilities would also have been correct e.g. 4 next to 2 and 3 next to 1

ACTIVITY RELATIONSHIP CHART

Plant TRESISA Project A-35
 Charted by JT With _____
 Date 1/14 Sheet 1 of 1
 Reference 35

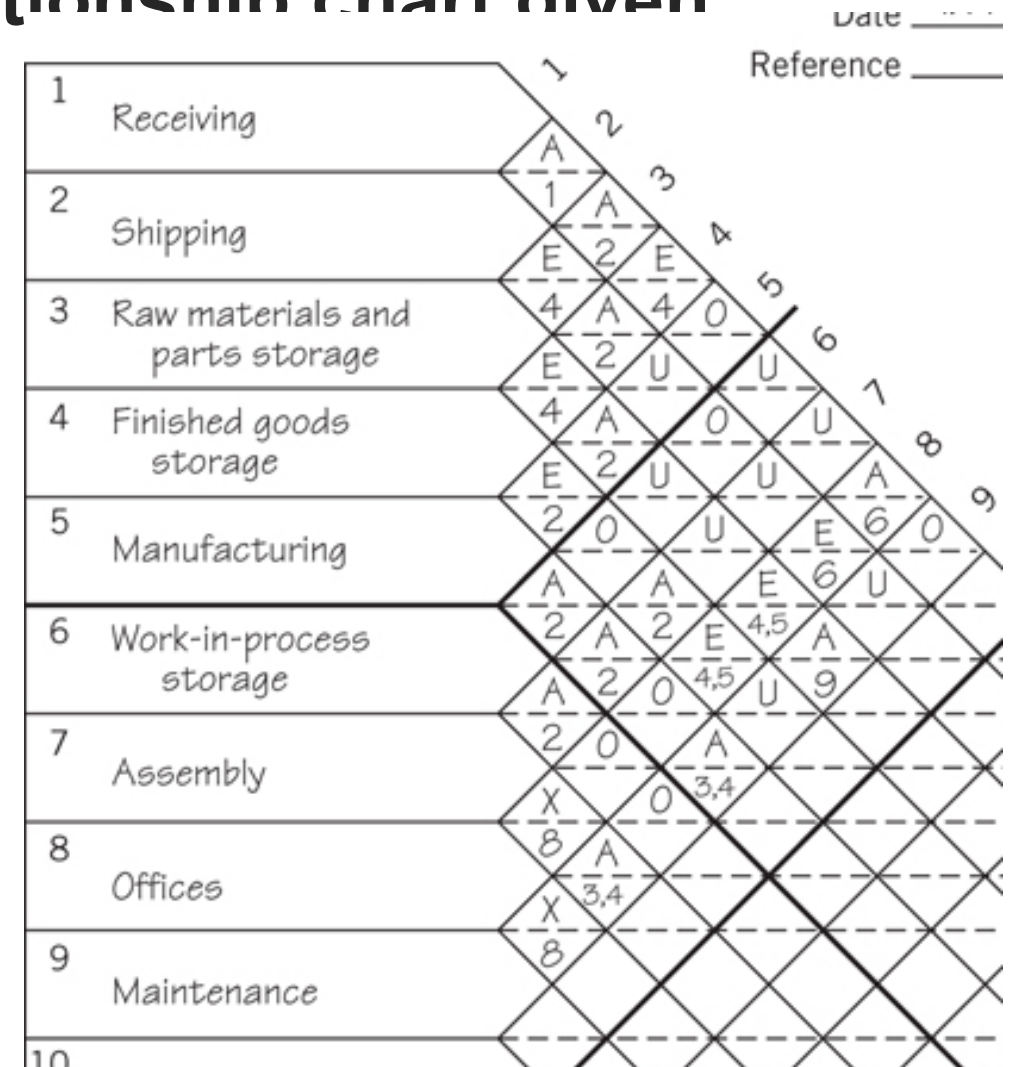


**Activity
relationship
chart**

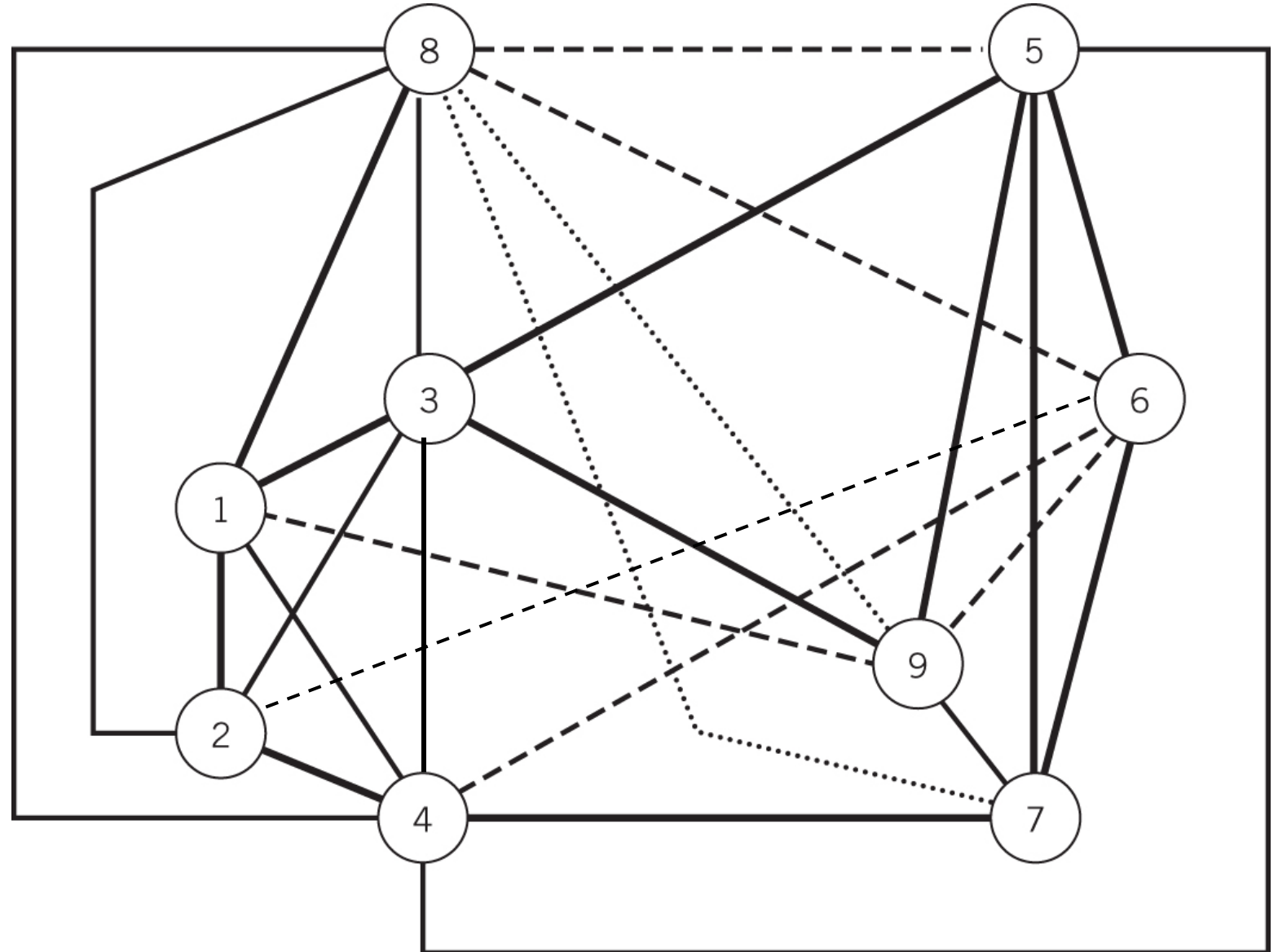
**SLP – example2 p300
to illustrate SLP logic
with activity
relationship chart given**

REASONS =
Out of Scope

A – 10 **————**
 E – 7 **————**
 I – 0 **————**
 O – 7 **-----**
 U – 8 **no line**
 X – 2 **.....**



Resulting Relationship diagram



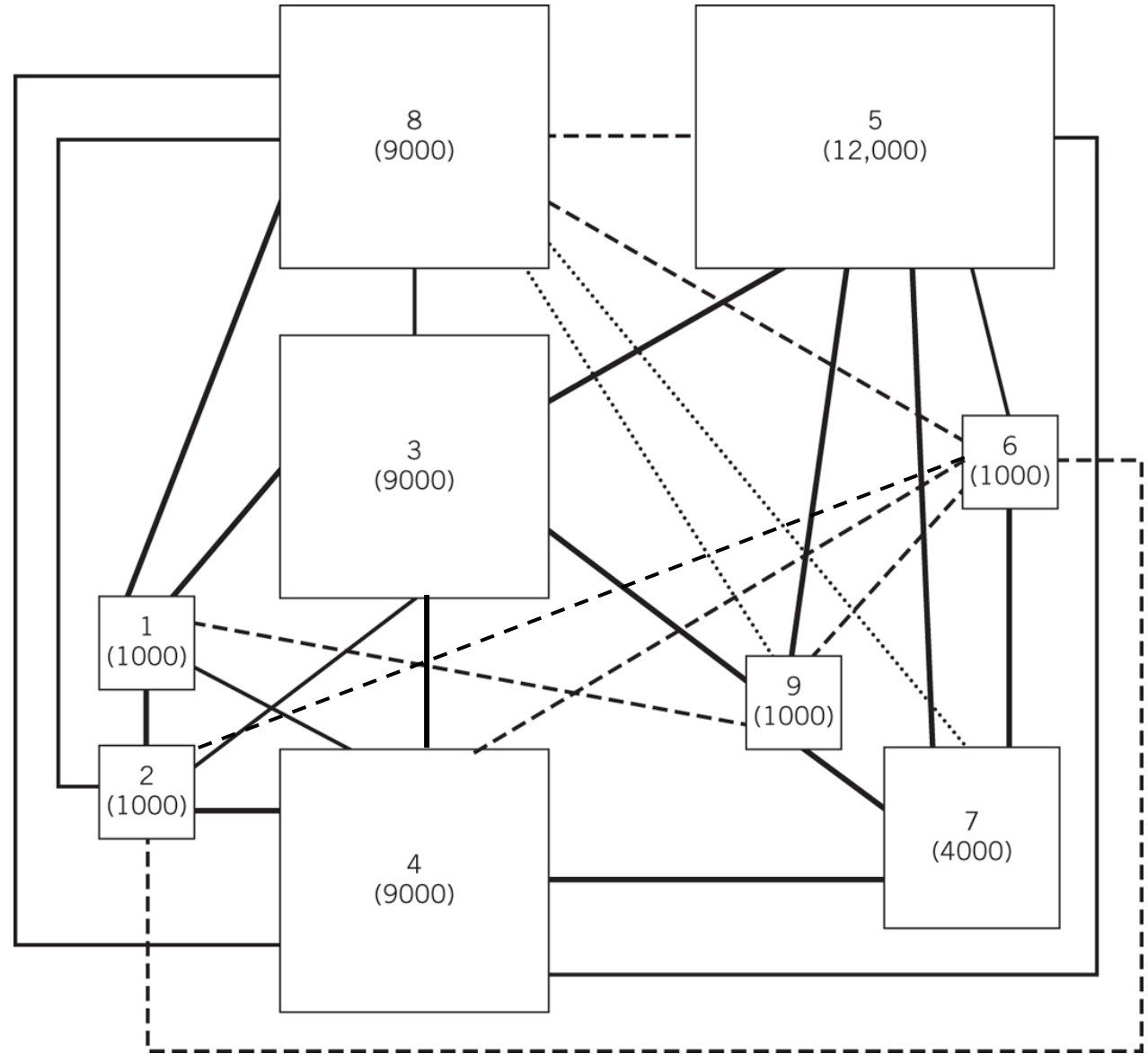
Resulting Space relationship diagram

(imagine rubber bands)

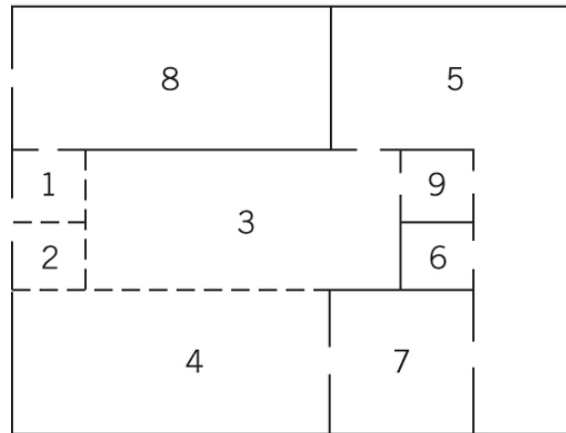
Where does one get department sizes?

From workstations footprints (refer Stephens Ch7 p163-167 x nr of workstations needed to make production demand)

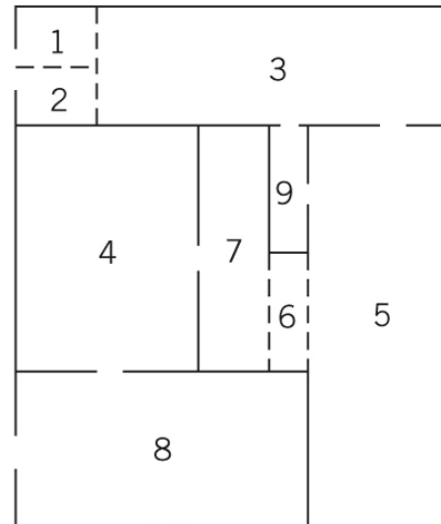
Block – squares or rectangles most easy to use, but you will have to ‘fit’ it into a building anyway



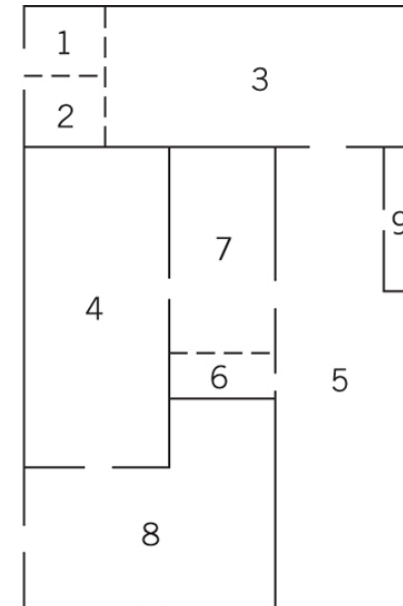
Resulting Alternative Block Layouts 9 departments



(a)



(b)



(c)

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Can you see the logic from the space relationship diagram, but then also the creative side? Intuition, judgment and experience help in this process.

From-To chart

Department Name	Area (ft ²)	No. of Grids	Flow							
			A	B	C	D	E	F	G	H
1. A: Receiving	12,000	30	0	45	15	25	10	5	0	0
2. B: Milling	8000	20	0	0	0	30	25	15	0	0
3. C: Press	6000	15	0	0	0	0	5	10	0	0
4. D: Screw m/c	12,000	30	0	20	0	0	35	0	0	0
5. E: Assembly	8000	20	0	0	0	0	0	65	35	0
6. F: Plating	12,000	30	0	5	0	0	25	0	65	0
7. G: Shipping	12,000	30	0	0	0	0	0	0	0	0
8. H: Dummy	2000	5	0	0	0	0	0	0	0	0

table_06_03

SLP – example 6.1 starting from From-To chart Ex 6.1-6.4

H = expansion space

Activity relationship chart

Ranking (or given, gut feel,
or sometimes calculation):
Max = 90 / 5 = intervals
0-18 = U
19-36 = O
37-54 = I
55-72 = E
73-90 = A

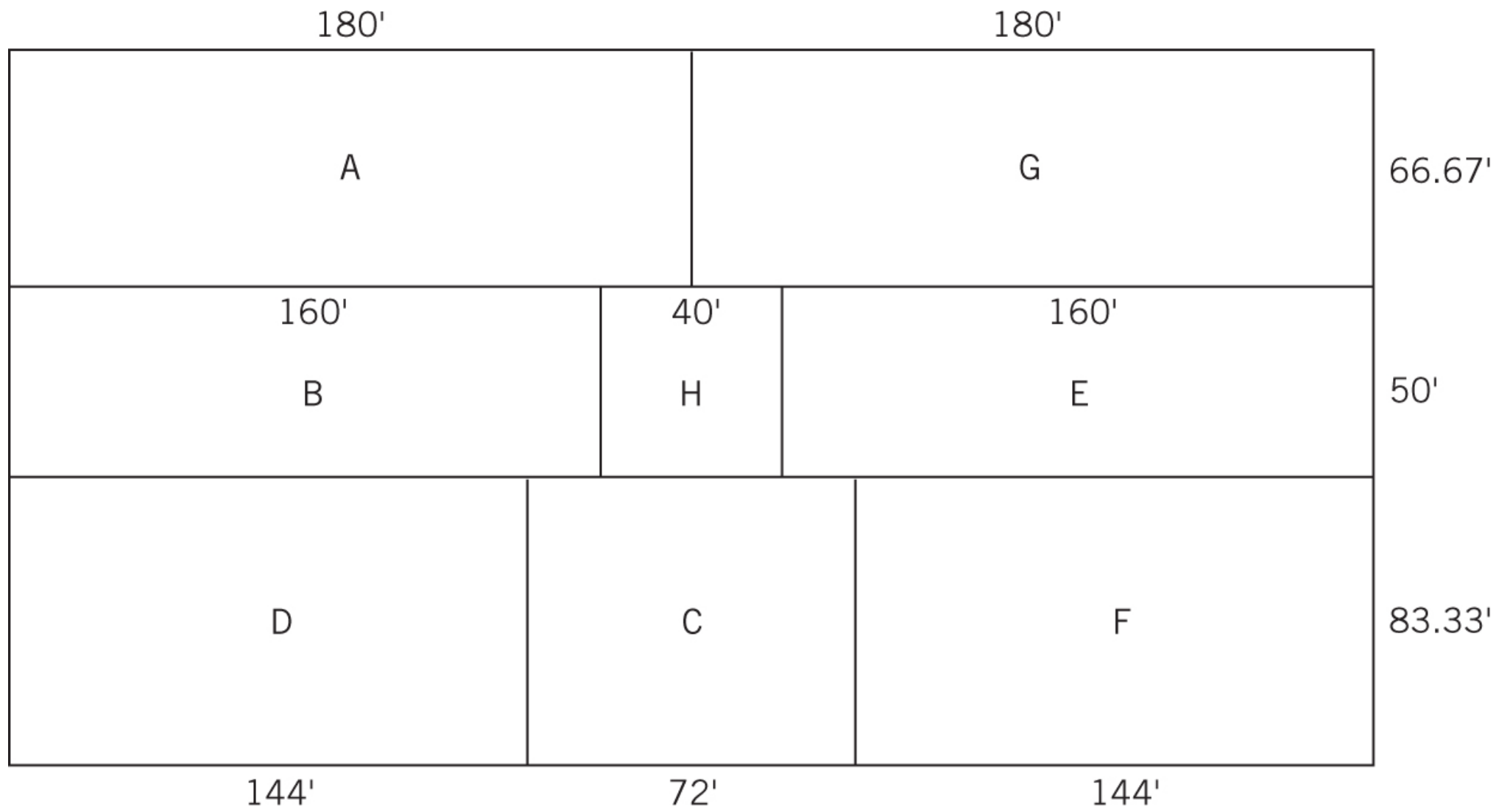
	A	B	C	D	E	F	G	H		A	B	C	D	E	F	G	H
A	0	45	15	25	10	5	0	0	A	—	I	U	O	U	U	U	U
B		0	0	50	25	20	0	0	B		—	U	I	O	O	U	U
C			0	0	5	10	0	0	C			—	U	U	U	U	U
D				0	35	0	0	0	D				—	O	U	U	U
E					0	90	35	0	E					—	A	O	U
F						0	65	0	F						—	E	U
G							0	0	G							—	U
H								0	H								—

(a) Flow-between chart

(b) Relationship chart

table_06_04

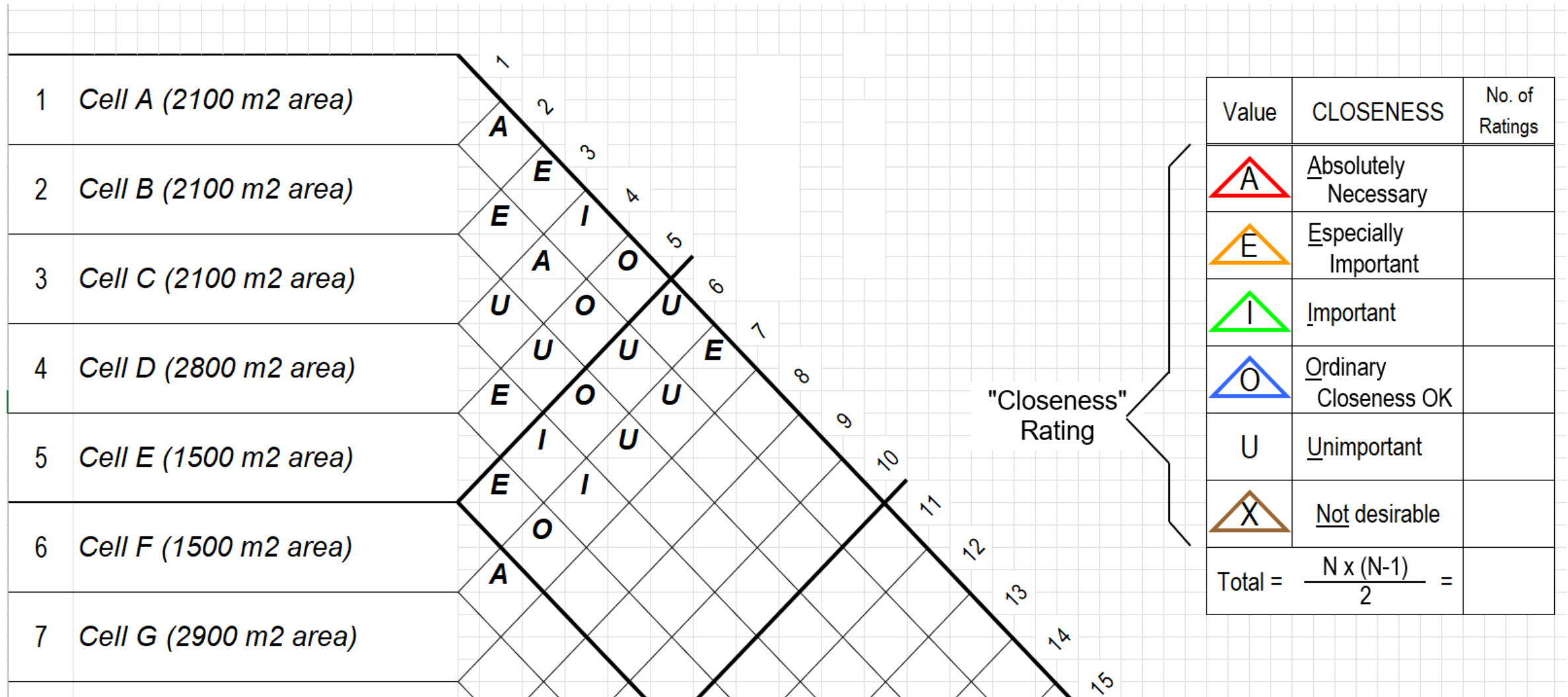
SLP – resulting block layout 8 departments



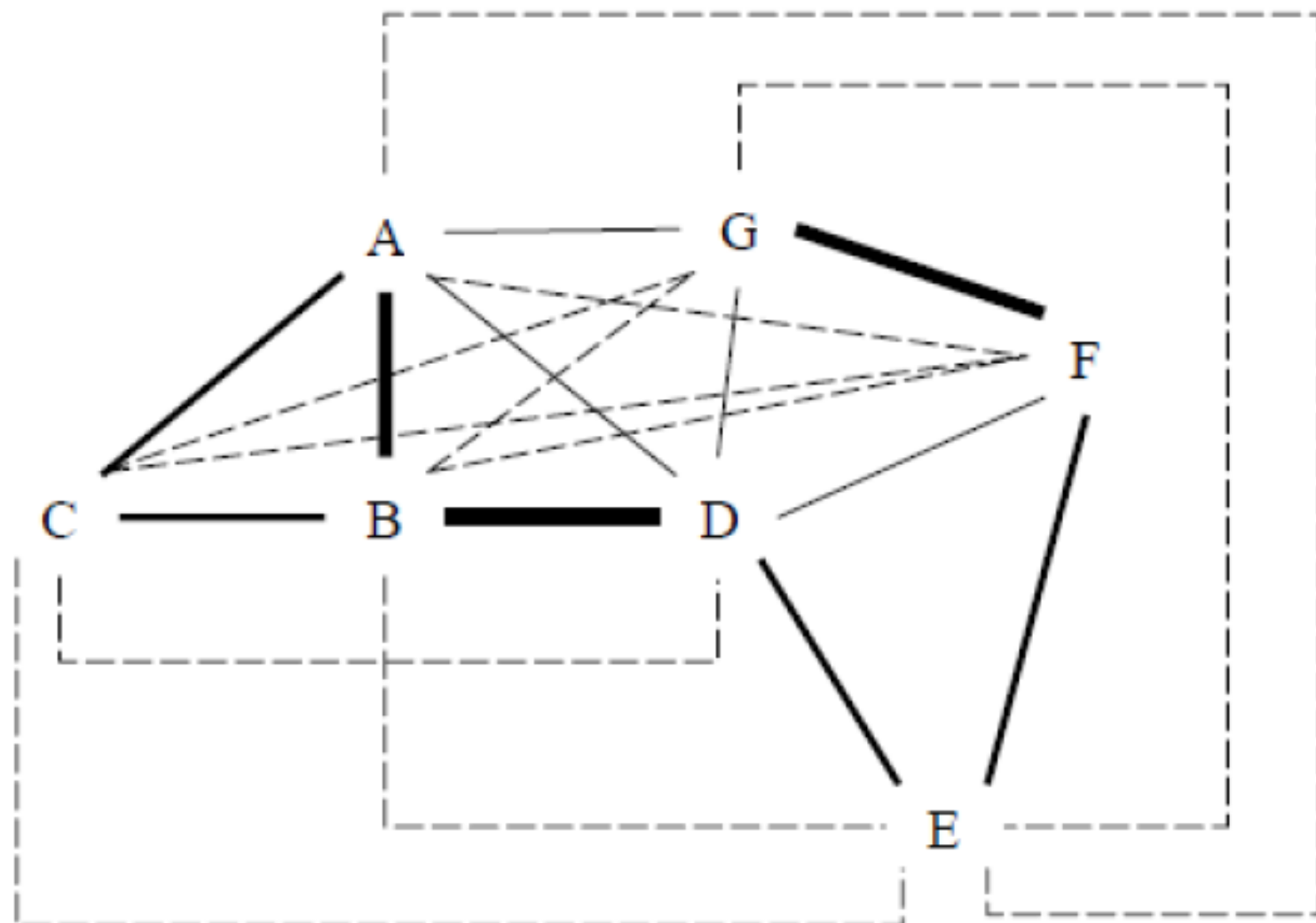
SLP – further examples....

Let us work through problem 6.10 in Tompkins book – 7 departments,
different sizes

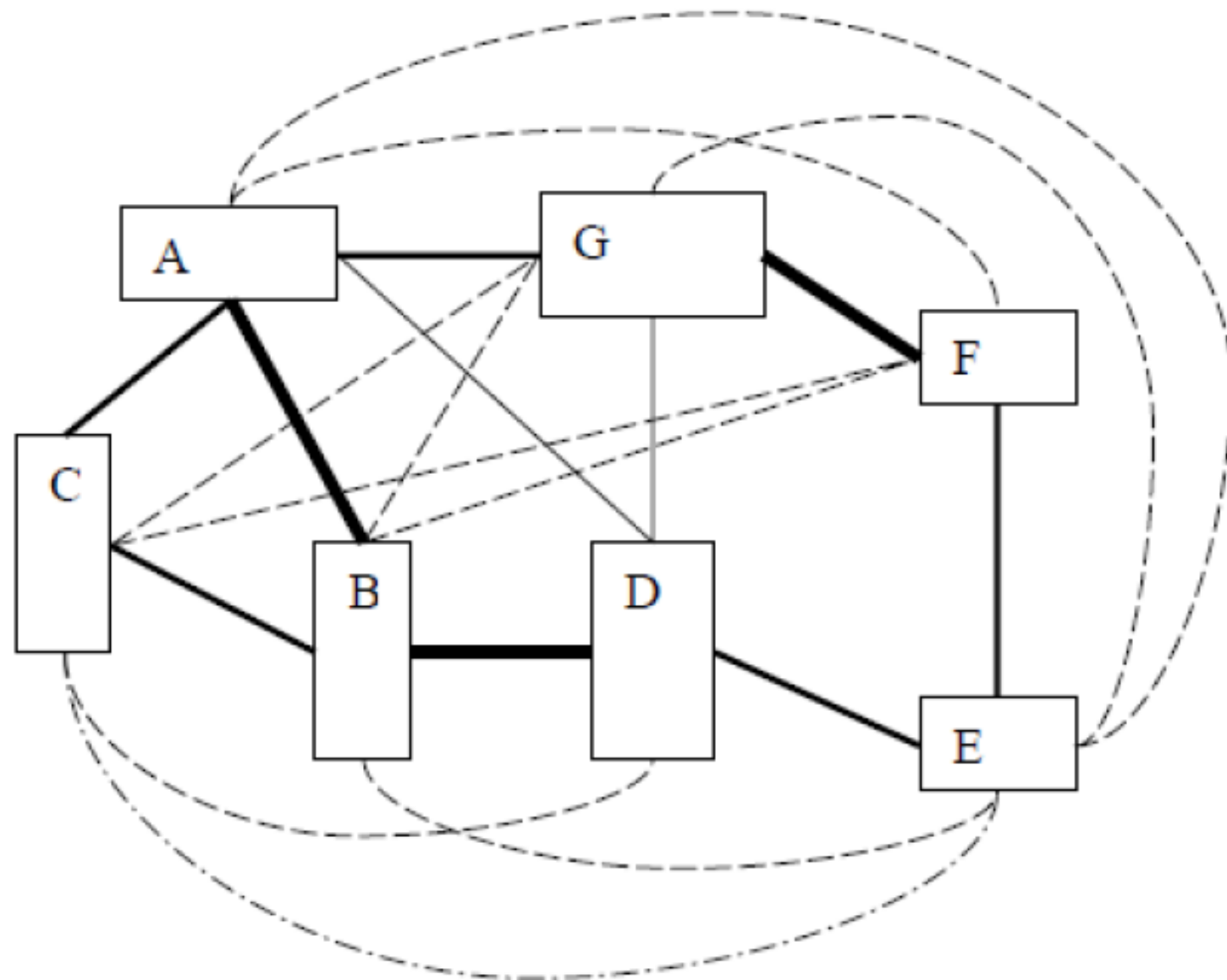
Problem 6.10 – 7 departments



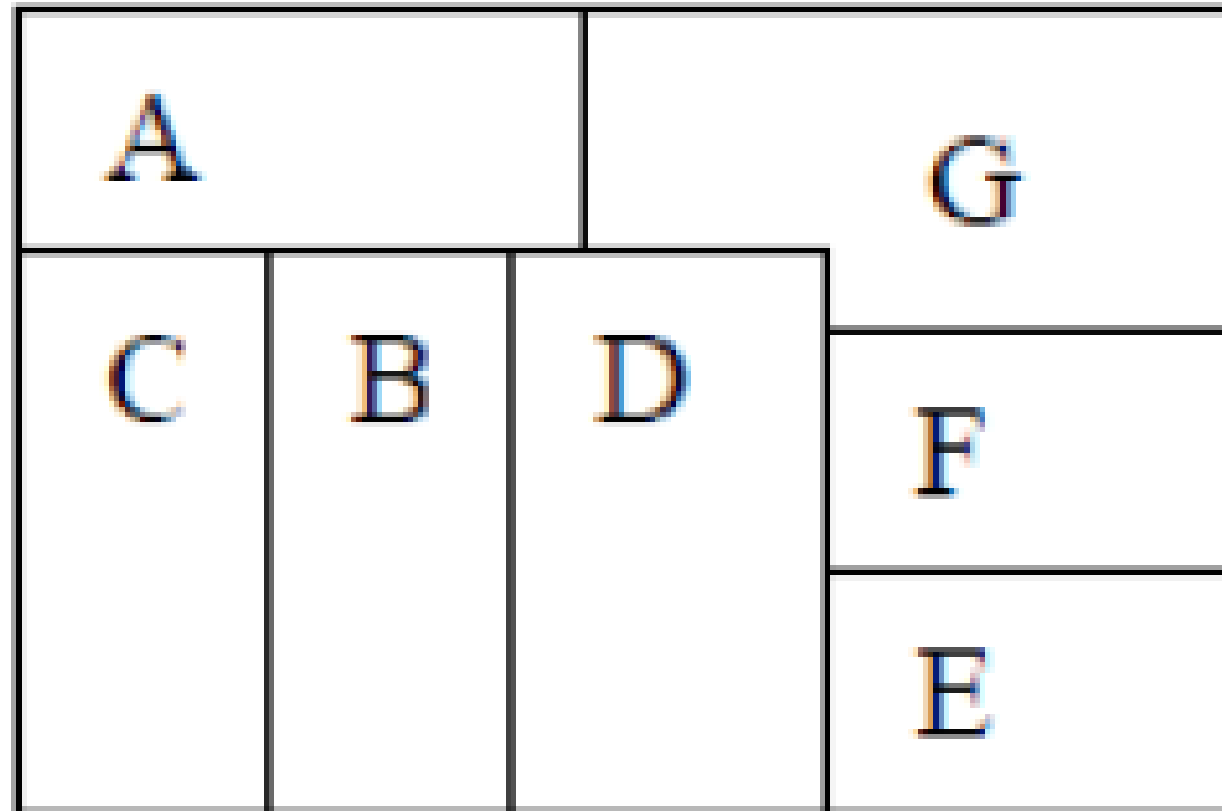
6.10a Relationship Diagram



6.10 Space Relationship Diagram



6.10b Block Layout using SLP



Other learning materials on SLP....

Muther himself: <https://www.youtube.com/watch?v=3orNXGODkis>

<https://www.youtube.com/watch?v=9xIA--aZoSM>

www.richardmuther.com for free working forms.(RMA130)